Question	Answer	Marks
10(a)(i)	cannot predict when a particular nucleus will decay **or** cannot predict which nucleus will decay next	**B1**
10(a)(ii)	(decay is) not affected by external (environmental) factors	**B1**
10(a)(iii)	fluctuations in (measured) count rate	**B1**
10(b)(i)	• large nuclei undergo fission whereas small nuclei undergo fusion • fission involves one nucleus splitting into two (or more) (smaller) nuclei • fusion involves two nuclei joining together to form one (larger) nucleus • fission is (usually) initiated by neutron bombardment • fusion is (usually) initiated by (very) high temperatures *Any three points, 1 mark each*	**B3**
10(b)(ii)	binding energy per nucleon is greatest for intermediate nucleon numbers (may be shown on sketch graph with axes labelled 'binding energy per nucleon' and 'nucleon number')	**B1**
	both fusion and fission involve an increase in binding energy (per nucleon)	**B1**